

ARKA, Russian Federation

WMO: 249880

Lat: 60.083N

Lon: 142.333E

Elev: 198

StdP: 98.97

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -40.6	(c) -38.5	(d) -43.4	(e) 0.1	(f) -40.5	(g) -41.3	(h) 0.1	(i) -38.3	(j) 6.5	(k) -11.4	(l) 6.0	(m) -11.0	(n) 0.4	(o) 10	(p) 0.719

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.2	(c) 26.3	(d) 16.5	(e) 24.0	(f) 15.5	(g) 21.9	(h) 14.6	(i) 17.7	(j) 23.9	(k) 16.4	(l) 22.1	(m) 15.4	(n) 20.1	(o) 2.0	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.5	11.2	19.6	14.3	10.4	17.5	13.4	9.8	16.6	51.1	24.0	46.9	22.6	43.9	20.1	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5.5	4.6	4.0	-43.2	30.9	2.3	2.2	-44.8	32.5	-46.2	33.8	-47.5	35.1	-49.1	36.7
			-43.1	20.1	2.1	1.8	-44.6	21.4	-45.8	22.5	-47.0	23.6	-48.5	24.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.9	-25.0	-21.9	-14.6	-4.5	4.6	11.5	14.6	14.0	7.0	-4.3	-17.2	-23.7
	DBStd	15.55	8.00	6.84	5.97	4.86	3.44	3.63	2.83	2.59	3.73	5.67	7.56	7.50
	HDD10.0	5784	1086	894	761	436	172	26	1	3	102	443	815	1046
	HDD18.3	8477	1344	1127	1019	686	427	207	121	136	339	701	1065	1304
	CDD10.0	356	0	0	0	0	3	69	143	127	13	0	0	0
	CDD18.3	7	0	0	0	0	0	1	5	2	0	0	0	0
	CDH23.3	255	0	0	0	0	3	81	124	47	1	0	0	0
	CDH26.7	53	0	0	0	0	1	18	28	6	0	0	0	0
Wind	WSAvg	1.6	1.7	1.6	1.6	1.6	1.7	1.7	1.4	1.3	1.4	1.7	1.9	1.9
Precipitation	PrecAvg	446	9	7	9	26	41	51	77	74	69	43	23	16
	PrecMax	895	57	28	40	68	119	113	368	218	168	114	88	74
	PrecMin	255	0	0	0	0	2	5	16	12	17	2	0	0
	PrecStd	120	13	8	12	17	26	26	62	45	41	28	23	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.9	-2.5	2.7	9.2	22.0	29.6	30.3	27.2	20.8	10.0	1.9	-3.0
		MCWB	-5.1	-4.4	-2.0	2.8	11.2	16.9	18.6	17.4	12.7	4.5	0.2	-5.7
	2%	DB	-5.8	-4.7	-0.2	7.0	17.9	25.7	26.7	24.4	17.9	7.0	-0.3	-5.7
		MCWB	-7.3	-6.7	-3.0	1.4	9.0	15.1	17.6	16.0	10.6	2.8	-1.5	-7.4
	5%	DB	-8.4	-7.3	-2.4	5.1	15.3	23.2	24.3	22.5	15.7	4.8	-2.3	-8.3
		MCWB	-9.8	-9.1	-5.0	0.4	7.6	14.1	16.3	15.4	10.0	1.5	-3.4	-9.9
	10%	DB	-11.5	-10.1	-4.1	3.3	12.8	20.4	21.8	20.3	13.8	3.1	-5.0	-11.2
		MCWB	-12.7	-11.7	-6.5	-0.7	6.4	12.9	15.1	14.6	9.8	0.3	-6.1	-12.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.4	-4.0	-0.9	3.2	11.6	17.6	20.7	19.1	14.0	5.7	0.8	-5.2
		MCDB	-3.0	-2.6	0.9	8.3	21.1	26.5	28.9	23.7	17.8	8.3	1.7	-3.4
	2%	WB	-7.2	-6.7	-3.0	1.7	9.6	15.9	18.5	17.3	12.8	3.6	-1.1	-7.4
		MCDB	-5.8	-4.8	-0.2	6.0	17.1	24.2	24.9	22.5	15.5	6.4	-0.3	-5.7
	5%	WB	-9.9	-9.2	-4.8	0.6	7.9	14.6	16.9	16.1	11.7	2.0	-3.5	-9.9
		MCDB	-8.4	-7.4	-2.4	4.2	14.2	21.8	22.8	20.6	14.1	3.9	-2.4	-8.4
	10%	WB	-12.7	-11.7	-6.5	-0.3	6.6	13.3	15.7	15.2	10.5	0.9	-6.1	-12.5
		MCDB	-11.4	-10.1	-4.2	2.7	12.4	19.5	20.7	18.9	12.8	2.4	-5.0	-11.3
Mean Daily Temperature Range	5% DB	MDBR	10.5	14.3	16.0	11.9	11.4	12.0	10.2	10.3	11.0	9.8	9.3	8.7
		MCDBR	7.8	10.2	15.4	13.3	17.9	19.7	17.3	16.1	14.9	11.0	5.4	7.8
		MCWBR	7.0	8.9	12.4	8.3	9.6	9.5	8.5	8.2	8.3	7.0	4.7	6.9
	5% WB	MCDBR	8.0	9.8	13.9	11.7	16.7	17.2	15.3	13.1	10.7	9.2	4.6	7.5
		MCWBR	7.2	8.6	11.2	7.5	9.3	8.9	8.1	7.4	6.4	6.2	4.3	6.8
Clear-Sky Solar Irradiance	taub		0.219	0.265	0.325	0.440	0.459	0.426	0.471	0.426	0.340	0.300	0.239	0.196
	taud		2.059	2.099	1.983	1.724	1.882	2.215	2.105	2.223	2.434	2.234	2.048	1.962
	Ebn at Noon		564	729	777	730	759	805	750	758	785	677	506	425
	Edh at Noon		43	76	120	190	175	128	140	116	79	67	43	30
All-Sky Solar Radiation	RadAvg		0.48	1.52	3.34	5.07	5.24	5.33	4.67	3.80	2.69	1.59	0.67	0.25
	RadStd		0.03	0.08	0.15	0.33	0.52	0.50	0.69	0.45	0.36	0.16	0.07	0.02

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page